

AL JOUF, Saudi Arabia

WMO: 403610

Lat: 29.785N

Lon: 40.100E

Elev: 2261

StdP: 13.53

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 35.5	(c) 37.8	(d) 4.8	(e) 7.7	(f) 51.7	(g) 8.9	(h) 9.4	(i) 55.8	(j) 29.4	(k) 52.8	(l) 25.4	(m) 55.5	(n) 5.0	(o) 240	(p) 0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 26.5	(c) 111.3	(d) 66.2	(e) 109.2	(f) 65.9	(g) 107.0	(h) 65.2	(i) 68.4	(j) 104.6	(k) 67.1	(l) 103.8	(m) 65.8	(n) 102.3	(o) 9.0	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.4	76.6	70.9	55.4	71.0	70.7	53.4	65.8	72.8	34.0	104.9	32.8	103.4	31.8	102.3	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	21.5	19.1		29.0	114.3	3.7	1.3	26.4	115.3	24.2	116.1	22.2	116.8	19.5	117.8
			DB	29.0	114.3	3.7	1.3	26.4	115.3	24.2	116.1	22.2	116.8	19.5	117.8
			WB	24.2	70.1	3.7	1.9	21.6	71.4	19.5	72.6	17.4	73.6	14.8	75.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.9	51.8	56.4	63.7	72.8	82.4	89.2	92.3	93.1	87.9	78.3	63.7	54.5		
	DBStd	15.81	5.66	6.23	7.09	6.54	5.76	4.22	4.48	4.19	4.62	5.87	6.78	5.88		
	HDD50	79	43	12	1	0	0	0	0	0	0	0	2	21		
	HDD65	1215	409	249	111	11	0	0	0	0	0	1	105	329		
	CDD50	8820	99	191	427	685	1005	1177	1311	1335	1138	879	412	161		
	CDD65	4483	1	8	72	246	541	727	846	870	688	415	65	4		
	CDH74	66625	22	118	851	2812	7383	11443	14020	14412	10250	4713	554	47		
(13)	CDH80	42535	2	23	304	1271	4310	7567	9754	10119	6592	2437	152	4		
(14)	Wind	WSAvg	8.5	7.8	8.6	9.7	9.6	9.5	9.4	8.7	7.7	7.4	7.3	6.9		
(15) Precipitation	PrecAvg	2.0	0.5	0.3	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.2	0.3	0.4		
	PrecMax	3.8	1.3	1.1	1.0	0.6	0.5	0.1	0.0	0.1	0.6	1.1	2.4	1.6		
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	0.9	0.4	0.3	0.3	0.2	0.1	0.0	0.0	0.0	0.1	0.3	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.0	83.8	94.7	98.7	106.0	111.2	113.4	113.4	110.8	101.0	89.3	79.2		
		MCWB	54.2	56.2	58.1	60.8	64.1	65.0	67.2	66.8	65.8	62.9	58.7	54.8		
	2%	DB	71.2	77.3	87.6	94.8	102.3	107.8	111.2	111.4	107.2	98.2	84.4	73.6		
		MCWB	50.7	52.5	56.9	60.0	62.6	64.6	66.2	66.6	65.1	61.7	57.3	53.0		
	5%	DB	66.4	73.3	82.5	91.3	100.0	105.6	109.0	109.3	104.2	95.2	80.6	69.5		
		MCWB	50.0	51.5	54.9	58.7	61.8	64.0	65.7	66.4	64.2	61.0	56.2	51.5		
	10%	DB	62.8	69.4	78.5	87.7	96.8	102.6	106.0	106.8	101.9	91.8	76.6	65.9		
		MCWB	48.4	50.1	53.3	57.3	60.9	63.0	64.8	65.7	63.7	60.2	55.2	50.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	58.4	61.2	63.8	66.5	68.2	70.1	70.8	69.0	65.7	62.9	59.0		
		MCDB	66.5	73.7	80.8	85.1	96.8	104.7	106.7	105.7	102.2	92.1	75.4	68.9		
	2%	WB	54.4	55.4	58.7	61.8	64.3	66.4	67.8	68.9	66.3	63.7	60.7	56.0		
		MCDB	65.7	70.7	80.3	86.5	94.8	103.4	105.7	104.7	100.7	89.2	72.2	68.3		
	5%	WB	51.6	53.3	56.4	60.2	62.9	64.8	66.4	67.4	65.1	62.4	59.0	53.6		
		MCDB	63.9	67.9	79.1	85.3	94.7	102.4	104.1	104.7	99.7	88.2	72.9	66.0		
	10%	WB	49.3	51.6	54.4	58.6	61.7	63.6	65.3	66.1	64.1	61.3	57.3	51.7		
		MCDB	61.2	66.3	75.9	83.9	93.3	100.7	102.6	104.1	98.5	86.9	72.7	63.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	22.1	23.6	24.4	24.6	25.9	25.7	26.5	26.3	24.4	21.6	21.1		
		MCDBR	25.2	27.5	28.7	27.9	27.4	27.9	27.1	27.3	27.8	27.0	25.6	25.1		
		MCWBR	12.6	13.0	12.8	11.6	10.4	10.5	9.8	10.2	10.4	10.6	10.9	12.6		
		MCDBR	22.2	23.4	25.7	25.2	25.3	26.9	26.1	26.7	26.5	24.1	20.6	20.8		
	5% WB	MCWBR	12.0	12.0	12.4	11.2	9.6	10.3	9.8	10.7	10.2	9.3	8.9	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.344	0.380	0.434	0.516	0.520	0.456	0.441	0.463	0.438	0.462	0.402	0.362		
	taud		2.186	2.062	1.907	1.734	1.746	1.909	1.979	1.922	1.962	1.892	2.023	2.125		
	Ebn at Noon		278	277	269	251	250	266	269	262	264	246	252	263		
	Edh at Noon		39	48	60	74	73	62	58	60	56	56	45	39		
(44) All-Sky Solar Radiation	RadAvg		1223	1536	1903	2183	2403	2626	2536	2341	2046	1628	1278	1127		
(45)	RadStd		62	74	63	69	100	49	68	53	57	62	87	39		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+2.50	N/A	N/A	+1.98	N/A	N/A	-58	-257	+832	+551
(47) Regional (1 neighbor)	+2.50	N/A	N/A	+1.98	N/A	N/A	-58	-257	+832	+551

Nomenclature: See separate page